Smart IoT Door Lock with Emergency Alert System

Project Type

System Design and Prototype Plan

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Project Overview:

- Designed a smart IoT-based door lock monitoring system to enhance residential security.
- The system records every **lock and unlock event** and reports the activity with timestamps to a **web or mobile interface**.
- Detects **unauthorized key insertion or suspicious access attempts** and triggers alert notifications.
- Includes a **portable emergency remote button** that allows users to send instant alerts during unsafe situations.
- Maintains **activity logs of door access events** for monitoring and security awareness.

Technologies (Planned)

- IoT
- ESP32 / Arduino
- Sensors
- Web Application
- Notification System

Acknowledgement:

- This document was prepared with the **assistance of AI tools** for writing support, formatting, and background research.
- The **project idea, problem identification, and proposed system design** were conceptualized by me (i.e, Vulli Thraiambica - Author)
- As a beginner exploring IoT systems and system design, the author used AI tools to support learning, improve clarity in documentation, and better understand technical concepts during the research process.
- AI tools were used only to **organize information, refine explanations, and assist with documentation**, while the core concept and intended solution remain the author's own.

Problem Statement:

- Traditional door lock systems provide **only physical security** and do not offer any digital monitoring or alert mechanisms.
- Homeowners usually **cannot track when the door is locked or unlocked**, especially when they are away from home.
- In cases of **unauthorized key insertion, forced access, or suspicious activity**, there is no automatic system to notify the homeowner.
- During emergency situations, occupants may **not have a quick and simple way to alert nearby people or trusted contacts for help**.
- Existing security solutions often require **expensive smart lock systems or complex installations**, making them less accessible for many households.

Therefore, there is a need for a **smart, affordable, and easy-to-monitor door lock system** that can:

- Track door lock activity in real time
- Detect unauthorized access attempts
- Send alert notifications during suspicious activity
- Provide an emergency alert option for users

The proposed **Smart IoT Door Lock with Emergency Alert System** aims to address these issues by integrating IoT technology, sensors, and a monitoring interface.

Proposed Solution:

The proposed solution is a **Smart IoT Door Lock with Emergency Alert System** designed to improve home security by monitoring door access events and providing real-time alerts.

The system integrates **IoT hardware, sensors, and a monitoring platform** to track lock activity and detect suspicious access attempts.

Key aspects of the proposed solution include:

- The system records every **lock and unlock event** and sends the activity information with **timestamps** to a web or mobile interface.
- Sensors connected to a **microcontroller (ESP32 / Arduino)** detect door lock activity and communicate the information through an internet connection.
- If an **unauthorized key insertion or abnormal access attempt** is detected, the system automatically generates an **alert notification**.
- A **portable emergency remote button** allows users to instantly trigger an alert during unsafe situations, notifying the homeowner or trusted contacts.

- All access events such as **lock actions, unlock actions, and failed access attempts** are stored as **activity logs** for monitoring and security awareness.
- The monitoring interface allows users to **view door lock status and access history remotely**.

Through this design, the system aims to provide a **simple, affordable, and intelligent security solution** that enhances awareness and response during suspicious or emergency situations.

System Architecture:

The Smart IoT Door Lock system is designed using a **modular architecture** that integrates hardware components, communication infrastructure, and a monitoring interface.

The system consists of the following main components:

1. Smart Door Lock Unit

- The physical door lock is equipped with sensors to detect **lock and unlock actions**.
- The unit can also detect **unauthorized key insertion or tampering attempts**.
- These sensor signals are forwarded to the microcontroller for processing.

2. Microcontroller (ESP32 / Arduino)

- The microcontroller acts as the **central control unit of the system**.
- It receives input signals from the sensors connected to the lock system.
- It processes lock activity data and sends the information to the server through a network connection.

3. Communication Layer

- The system uses **Wi-Fi connectivity** to transmit data between the hardware unit and the backend server.
- Lock activity events are sent to the server in **real time**.

4. Backend Server / Cloud System

- The backend server receives and processes lock activity data.
- It stores **access logs with timestamps** in the database.
- If abnormal activity or unauthorized access is detected, the system generates **alert notifications**.

5. User Monitoring Interface

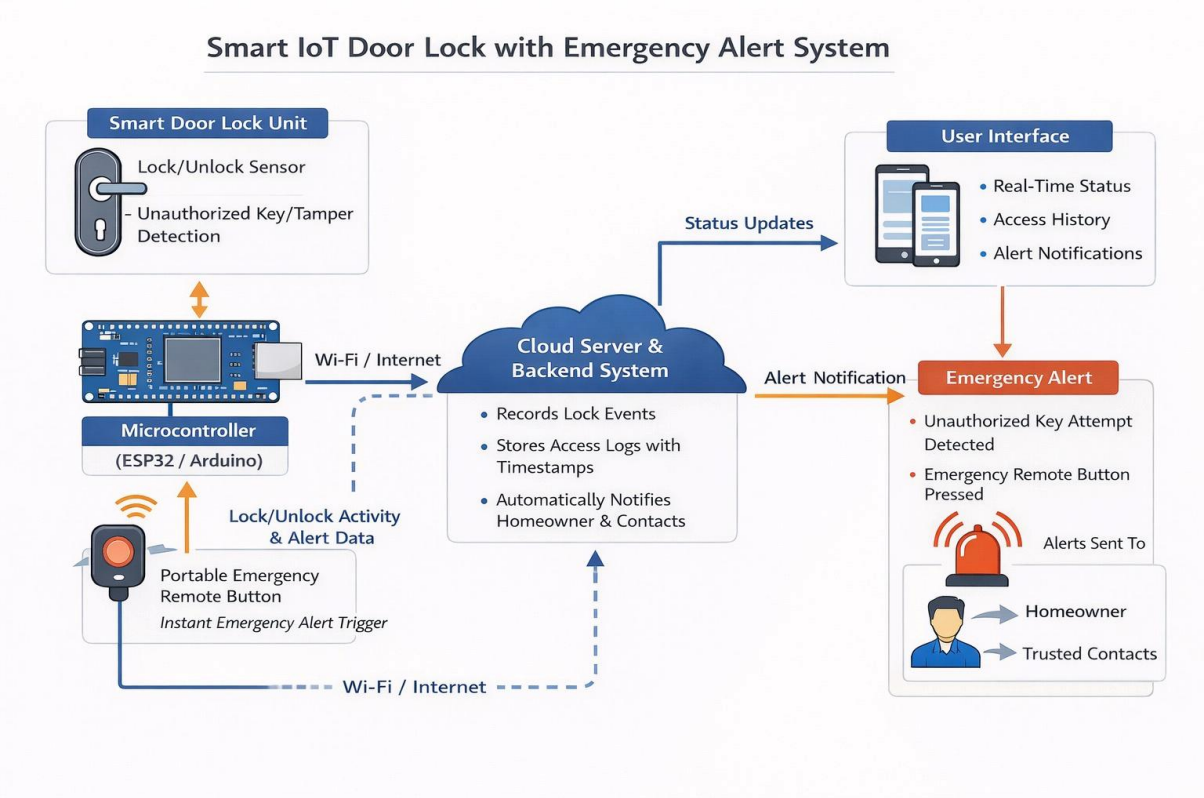
- Users can monitor the door lock through a **web or mobile interface**.
- The interface displays:
    - Current lock status
    - Access activity history

- Alert notifications

6. Emergency Remote Button

- The system includes a **portable emergency remote button**.
- When pressed, it sends an emergency signal to the system.
- The system then triggers **immediate alert notifications** to the homeowner or trusted contacts.

Architecture Diagram :



Complete System Workflow:

START

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System Powered ON

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Initialize System Components

- **Lock / Unlock Sensors**
- **Microcontroller (ESP32 / Arduino)**
- **Wi-Fi / Internet Communication**
- **Backend / Cloud Server**
- **Emergency Remote Button**

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System enters Monitoring Mode

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Door Lock Interaction Detected

(User inserts key or operates lock)

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Lock / Unlock Sensors Detect Activity

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Sensor Data Sent to Microcontroller

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Microcontroller Processes Event

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Determine Event Type

Is Access Authorized?

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YES – NORMAL ACCESS

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Lock / Unlock Event Detected

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Send Event Data to Backend Server

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Server Stores Event in Database

- Lock Status
- Timestamp
- Event Type

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Update Lock Status in Web / Mobile Interface

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User Can View Door Activity Log

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Return to Monitoring Mode

NO-UNAUTHORISED ACCESS

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Unauthorized Key Attempt Detected

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Microcontroller Flags Suspicious Activity

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Alert Data Sent to Backend Server

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Server Generates Alert

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Notification Sent to User / Homeowner

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Unauthorized Attempt Stored in Activity Log

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User Can View Alert in Monitoring Interface

CONTINUE SYSTEM MONITORING

Emergency Situation Flow

Portable Emergency Remote Button Pressed

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Emergency Signal Sent to Microcontroller

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Microcontroller Identifies Emergency Event

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Emergency Alert Sent to Backend Server

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Server Triggers Emergency Notification

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Alert Sent To

- **Homeowner**
- **Trusted Contacts / Neighbors**

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Emergency Event Stored in System Activity Log

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Users Receive Immediate Alert

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System Continues Monitoring

Final System Outcome

System Continuously Monitors Door Lock Activity

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All Events Stored in Activity Log

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Real-Time Notifications Generated When Required

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Users Monitor Lock Status and Access History

through Web / Mobile Interface

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END

System Workflow

The Smart IoT Door Lock system follows a structured workflow to monitor door access activity and generate alerts when necessary**.**

Normal Access Workflow

- A user interacts with the door lock by locking or unlocking the door using a key.
- Sensors attached to the lock system detect the lock or unlock action.
- The sensor data is sent to the microcontroller (ESP32 / Arduino).
- The microcontroller processes the input and identifies the lock activity event.
- The event information is transmitted through Wi-Fi to the backend server.
- The backend server records the event details and timestamp in the activity log database.
- The monitoring application updates the lock status and access history in real time.

Unauthorized Access Detection Workflow:

- If an unauthorized key insertion or abnormal access attempt is detected by the sensors, the microcontroller identifies it as a suspicious event.
- The event information is immediately sent to the backend server.
- The system generates an alert notification to the homeowner or registered user.

- The unauthorized access attempt is recorded in the system activity logs.

Emergency Alert Workflow:

1. In case of an unsafe situation, the user presses the portable emergency remote button.
2. The emergency signal is received by the microcontroller.
3. The system immediately sends an emergency alert notification to the homeowner or trusted contacts.

The emergency event is also stored in the system activity log

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System Explanation:

- The **Smart IoT Door Lock with Emergency Alert System** is designed to enhance residential security by integrating door lock monitoring, unauthorized access detection, and emergency alert functionality through IoT technology.
- The system operates using a combination of **hardware components, communication infrastructure, and a monitoring interface**.
- The **door lock unit** is equipped with sensors that detect lock and unlock actions whenever a user interacts with the door using a key. These sensors capture the lock activity and send the data to the **microcontroller (ESP32 / Arduino)**.
- The microcontroller acts as the **central processing unit of the system**. It processes the sensor input and determines whether the detected event corresponds to a normal access event or a suspicious activity such as an unauthorized key attempt.
- Once the event is processed, the microcontroller transmits the information through **Wi-Fi connectivity** to the **backend server or cloud system**. The server records all events including lock actions, unlock actions, unauthorized access attempts, and emergency alerts.
- The server stores this information in an **activity log database**, which maintains records of door access events along with timestamps. This allows users to track door activity and monitor security events.
- Users can access this information through a **web or mobile monitoring interface**. The interface allows users to view the current lock status, access history, and any alert notifications generated by the system.
- The system also includes a **portable emergency remote button** designed for situations where immediate assistance is required. When the button is pressed, the signal is sent to the microcontroller, which triggers an emergency alert through the backend server. The system then sends alert notifications to the homeowner or trusted contacts.
- By combining **real-time monitoring, event logging, and alert notifications**, the proposed system aims to provide a **simple and effective smart security solution** that improves awareness and response during suspicious or emergency situations.

Prototype Implementation Plan

The Smart IoT Door Lock system can be developed through a structured prototype implementation process. The following stages describe the planned development approach for building and testing the system.

Phase 1 — Hardware Setup

The initial stage focuses on building the hardware unit of the system.

- Install the **microcontroller (ESP32 / Arduino)** to act as the main processing unit.
- Connect **lock / unlock sensors** to detect door lock activity.
- Configure the **portable emergency remote button** to send emergency signals to the system.
- Ensure the microcontroller is capable of communicating with the network using **Wi-Fi connectivity**.

Phase 2 — Event Detection and Processing

In this stage, the system will be programmed to detect door activity.

- Sensors monitor **lock and unlock actions**.
- The microcontroller processes the sensor signals to determine the type of event.
- The system identifies whether the event corresponds to:
    - Normal access
    - Unauthorized access attempt
    - Emergency alert trigger.

Phase 3 — Server and Data Logging

The next stage involves setting up the backend system.

- Develop a **backend server** to receive event data from the microcontroller.
- Store event information such as **lock status, timestamps, and event type** in a database.
- Maintain an **activity log** that records all system events.

Phase 4 — Monitoring Interface

A monitoring interface will be developed for users.

- Create a **web or mobile dashboard** to display lock activity.
- Allow users to view:
  - Current lock status
  - Door access history
  - Alert notifications.

Phase 5 — Alert and Notification System

The final stage integrates the alert mechanism.

- Configure the system to send **notifications when unauthorized access is detected**.
- Implement **emergency alerts triggered by the remote button**.
- Ensure notifications reach the **homeowner or trusted contacts** in real time.

Through these stages, the prototype will demonstrate the functionality of the **Smart IoT Door Lock monitoring and emergency alert system**.

Key Features and Technologies

Key Features

The Smart IoT Door Lock with Emergency Alert System includes several features designed to improve home security and monitoring.

- **Real-Time Door Lock Monitoring**
  The system detects and records every **lock and unlock event**, allowing users to monitor door activity.
- **Activity Logging with Timestamp**
  All events are stored in a **database with timestamps**, enabling users to track door access history.
- **Unauthorized Access Detection**
  The system can detect **unauthorized key insertion or suspicious access attempts** and trigger alert notifications.
- **Emergency Alert Button**
  A **portable emergency remote button** allows users to instantly trigger an alert in unsafe situations.
- **Instant Alert Notifications**
  The system sends **real-time alerts to the homeowner or trusted contacts** when suspicious activity or emergency events occur.

- **Remote Monitoring Interface**
  Users can monitor **door lock status and activity logs through a web or mobile interface**.

Technologies (Planned):

The proposed system integrates the following technologies:

- **Internet of Things (IoT)**
  Enables communication between hardware components and the monitoring system.
- **Microcontroller (ESP32 / Arduino)**
  Acts as the central processing unit that receives sensor data and manages communication.
- **Sensors**
  Detect lock and unlock actions and help identify unauthorized access attempts.
- **Wi-Fi Communication**
  Used to transmit event data from the microcontroller to the backend server.
- **Backend Server and Database**
  Stores lock activity logs, processes events, and manages alert notifications.
- **Web / Mobile Monitoring Interface**
  Provides users with real-time access to door lock status and activity history.

Conclusion:

The **Smart IoT Door Lock with Emergency Alert System** is designed to enhance residential security by integrating real-time monitoring, unauthorized access detection, and emergency alert functionality into a single system.

The proposed system uses **sensors, a microcontroller, and internet connectivity** to monitor door lock activity and record events such as lock actions, unlock actions, and suspicious access attempts. These events are stored in a backend system and can be viewed through a **web or mobile monitoring interface**, allowing users to track door activity remotely.

In addition to monitoring capabilities, the system includes an **emergency alert feature through a portable remote button**, enabling users to quickly notify homeowners or trusted contacts during unsafe situations.

The system architecture, workflow design, and prototype implementation plan presented in this document demonstrate a **feasible approach for developing an IoT-based smart security solution**. By combining activity logging, alert notifications, and remote monitoring, the proposed design aims to improve awareness and response during potential security threats.

Overall, this project represents an effort to explore **practical applications of IoT technology in home security systems** while providing a foundation for further development and implementation in real-world environments.